

REQUEST 31 / LAST SIGNAL BEFORE DARK

ZOMBIE INVASION PREP / EMERGENCY RESPONSE EXERCISE

CLASSROOM SIMULATION • OPERATIONS REQUEST

SITUATION

The evacuation convoy is waiting for a route update. The relay radio has one battery left, and the road crew cannot hear headquarters. A technician has proposed a creek-powered charging station.

REQUEST

Plan a shallow diversion intake for a technician-operated microhydro unit. Decide whether it can sustain a 25 W charging demand.

SITE INFORMATION

A proposed U-shaped intake has a 12 ft front and two 6 ft returns. Fill staging is 180 ft away on a dry level path. An existing downhill pipe route provides 5 m of drop; flow is 2 L/s. Assume 40% overall efficiency, including losses. Power estimate: $\text{watts} = 9.81 \times \text{flow (L/s)} \times \text{drop (m)} \times \text{efficiency}$.

REQUESTED TIME WINDOW

Complete construction within 3 hours of assignment. Delivery, setup, inspection and finishing count. State weather/daylight assumptions and choose your planning factor.

OPERATIONAL CONSTRAINT

The drop comes from the terrain, not a tall sandbag dam. Request screened intake, pipe, turbine, generator and charging controls; a qualified adult handles installation. Show a safe bypass and a battery fallback.

ESTIMATE TO SUBMIT

Use the existing worksheet and your measured rate: bags and fill; time with 1 team and 4 teams; teams, people, equipment and worker support to meet the deadline. Choose and justify a counting rule; include water routing, access, and overflow or equipment-failure fallback. Engineering performance checks are an extension after the core estimate.

PLANNER: _____ **TEAM RATE:** _____

CLASSROOM EXERCISE — NOT A REAL DEPLOYMENT ORDER

REQUEST 32 / THE WHEEL THAT KEEPS TURNING

ZOMBIE INVASION PREP / EMERGENCY RESPONSE EXERCISE

CLASSROOM SIMULATION • OPERATIONS REQUEST

SITUATION

The fuel shed is empty. At the shelter garden, seed trays are drying while a small mechanical irrigation pump sits idle. An old demonstration water wheel could drive it through a guarded belt.

REQUEST

Plan the intake and channel for a technician-approved wheel system; keep the garden watered without an electric motor.

SITE INFORMATION

Proposed U intake: 8 ft front and two 4 ft returns. Two lined channel edges are each 15 ft long. Fill staging is 90 ft away. Use a supplied demonstration rating of 12 pump strokes/minute and 0.20 gallon/stroke. The garden needs 180 gallons.

REQUESTED TIME WINDOW

Complete construction within 4 hours of assignment. Delivery, setup, inspection and finishing count. State weather/daylight assumptions and choose your planning factor.

OPERATIONAL CONSTRAINT

Request wheel, frame, bearings, guarded drive, pump and liner. Sandbags shape water delivery; they are not wheel bearings or machine supports. Keep the bypass open and rotating parts inaccessible.

ESTIMATE TO SUBMIT

Use the existing worksheet and your measured rate: bags and fill; time with 1 team and 4 teams; teams, people, equipment and worker support to meet the deadline. Choose and justify a counting rule; include water routing, access, and overflow or equipment-failure fallback. Engineering performance checks are an extension after the core estimate.

PLANNER: _____ **TEAM RATE:** _____

CLASSROOM EXERCISE — NOT A REAL DEPLOYMENT ORDER

REQUEST 33 / WATER MUST CLIMB THE HILL

ZOMBIE INVASION PREP / EMERGENCY RESPONSE EXERCISE

CLASSROOM SIMULATION • OPERATIONS REQUEST

SITUATION

The hilltop food garden is the shelter's next harvest. Its irrigation tank is empty, the generator has failed, and carrying buckets up the hill will pull workers away from evacuation support.

REQUEST

Plan a shallow intake for a proper hydraulic ram pump and estimate whether it can deliver 120 gallons within four operating hours.

SITE INFORMATION

Proposed U intake: 10 ft front and two 5 ft returns. Fill staging is 240 ft away on level ground. Source flow through the drive pipe is 6 gal/min; fall to pump is 4 ft; lift from pump to tank is 24 ft. Assume efficiency 0.60. $\text{Delivery gal/min} = \text{efficiency} \times \text{source flow} \times \text{fall} \div \text{lift}$.

REQUESTED TIME WINDOW

Complete construction within 3 hours of assignment. Delivery, setup, inspection and finishing count. State weather/daylight assumptions and choose your planning factor.

OPERATIONAL CONSTRAINT

Construction window is separate from four operating hours. Request ram pump, drive pipe, delivery pipe, screened intake and a rated tank. Plan return drainage for water not pumped uphill; do not create an improvised elevated reservoir.

ESTIMATE TO SUBMIT

Use the existing worksheet and your measured rate: bags and fill; time with 1 team and 4 teams; teams, people, equipment and worker support to meet the deadline. Choose and justify a counting rule; include water routing, access, and overflow or equipment-failure fallback. Engineering performance checks are an extension after the core estimate.

PLANNER: _____ **TEAM RATE:** _____

CLASSROOM EXERCISE — NOT A REAL DEPLOYMENT ORDER

REQUEST 34 / THIRTY PEOPLE, ONE WORKING TOILET

ZOMBIE INVASION PREP / EMERGENCY RESPONSE EXERCISE

CLASSROOM SIMULATION • OPERATIONS REQUEST

SITUATION

The shelter toilets still work, but the water main is down. Thirty evacuees arrive before the next tanker. Rain is forecast, and staff want reserve flushing water ready before the roof begins shedding runoff.

REQUEST

Plan roof-runoff collection into a shallow lined basin for nonpotable flushing water.

SITE INFORMATION

Proposed basin inside dimensions: 12 × 8 ft; operating water depth 0.25 ft. Roof collection area is 600 ft². Forecast rainfall is 0.5 in; assume 80% collection efficiency. Expected runoff ft³ = roof area × rainfall in feet × efficiency. Demand: 30 people × 2 flushes × 1.5 gallons.

REQUESTED TIME WINDOW

Complete construction within 2 hours of assignment. Delivery, setup, inspection and finishing count. State weather/daylight assumptions and choose your planning factor.

OPERATIONAL CONSTRAINT

Ground-level site; fill staging 80 ft away. Request gutters, screen, liner and transfer containers. Keep overflow away from the shelter. No roof access by students; forecast rain is not water already in storage.

ESTIMATE TO SUBMIT

Use the existing worksheet and your measured rate: bags and fill; time with 1 team and 4 teams; teams, people, equipment and worker support to meet the deadline. Choose and justify a counting rule; include water routing, access, and overflow or equipment-failure fallback. Engineering performance checks are an extension after the core estimate.

PLANNER: _____ **TEAM RATE:** _____

CLASSROOM EXERCISE — NOT A REAL DEPLOYMENT ORDER

REQUEST 35 / CATCH THE RAIN BEFORE IT VANISHES

ZOMBIE INVASION PREP / EMERGENCY RESPONSE EXERCISE

CLASSROOM SIMULATION • OPERATIONS REQUEST

SITUATION

The shelter roof cannot be used: a fallen branch damaged the gutters. A passing shower may be the only chance to collect water for the garden before the heat returns.

REQUEST

Plan a supported tarp catchment feeding a shallow lined basin and keep the collection route from collapsing or overflowing.

SITE INFORMATION

Catchment tarp is 16 × 12 ft; supplied approved ground-edge anchoring plan uses 18 bag stations with 2 bags each. Basin inside dimensions: 8 × 6 ft at 0.25 ft operating depth. Forecast rainfall: 0.75 in; collection efficiency: 70%. Garden request: 50 gallons. Fill staging is 120 ft away.

REQUESTED TIME WINDOW

Complete construction within 3 hours of assignment. Delivery, setup, inspection and finishing count. State weather/daylight assumptions and choose your planning factor.

OPERATIONAL CONSTRAINT

Request suitable tarp supports, gutter/hose, screen and liner. The station count applies only to the supplied support plan; no improvised overhead water loads. Forecast yield = area × rainfall in feet × efficiency × 7.48.

ESTIMATE TO SUBMIT

Use the existing worksheet and your measured rate: bags and fill; time with 1 team and 4 teams; teams, people, equipment and worker support to meet the deadline. Choose and justify a counting rule; include water routing, access, and overflow or equipment-failure fallback. Engineering performance checks are an extension after the core estimate.

PLANNER: _____ **TEAM RATE:** _____

CLASSROOM EXERCISE — NOT A REAL DEPLOYMENT ORDER

REQUEST 36 / MUD IS KILLING THE PUMP

ZOMBIE INVASION PREP / EMERGENCY RESPONSE EXERCISE

CLASSROOM SIMULATION • OPERATIONS REQUEST

SITUATION

The shelter's cleanup pump is choking on grit. Volunteers keep clearing the intake, but muddy runoff returns faster than they can work. A second cleanout would stop the cleanup line.

REQUEST

Plan two shallow lined settling cells so one can be isolated and cleaned while the other remains available.

SITE INFORMATION

Each separate cell is 10 × 6 ft inside, with 0.25 ft operating depth. Cells have separate full perimeters. A further 12 ft two-layer line routes clean upslope rainwater around the station. Fill staging is 200 ft away. Assume an operating inflow of 3 gal/min.

REQUESTED TIME WINDOW

Complete construction within 4 hours of assignment. Delivery, setup, inspection and finishing count. State weather/daylight assumptions and choose your planning factor.

OPERATIONAL CONSTRAINT

Request two liners, controlled outlets, isolation valves, screens and sediment tools. Settling is pretreatment; transferred water remains nonpotable. Show isolated-cell drainage and an overflow route.

ESTIMATE TO SUBMIT

Use the existing worksheet and your measured rate: bags and fill; time with 1 team and 4 teams; teams, people, equipment and worker support to meet the deadline. Choose and justify a counting rule; include water routing, access, and overflow or equipment-failure fallback. Engineering performance checks are an extension after the core estimate.

PLANNER: _____ **TEAM RATE:** _____

CLASSROOM EXERCISE — NOT A REAL DEPLOYMENT ORDER

REQUEST 37 / THE SLICK AT THE DRAIN

ZOMBIE INVASION PREP / EMERGENCY RESPONSE EXERCISE

CLASSROOM SIMULATION • OPERATIONS REQUEST

SITUATION

A shiny slick is drifting toward the shelter's drainage outlet. Blocking all the water would flood the loading area; letting it all through could carry the slick downstream.

REQUEST

Prepare a paper plan for responder-controlled underflow containment: retain floating material while providing a submerged water outlet.

SITE INFORMATION

Proposed U containment footprint: 14 ft front and two 7 ft returns. Use a shallow 0.25 ft planning water depth and a two-wide, two-layer perimeter. Fill staging is 100 ft outside the exclusion zone. Hypothetical inflow is 8 gal/min; approved outlet rating is 10 gal/min.

REQUESTED TIME WINDOW

Complete construction within 2 hours of assignment. Delivery, setup, inspection and finishing count. State weather/daylight assumptions and choose your planning factor.

OPERATIONAL CONSTRAINT

Qualified responders specify liner, outlet geometry and recovery equipment. Do not invent an outlet opening from bag counts. Underflow does not purify water. A student demonstration uses only a contained tray and vegetable oil.

ESTIMATE TO SUBMIT

Use the existing worksheet and your measured rate: bags and fill; time with 1 team and 4 teams; teams, people, equipment and worker support to meet the deadline. Choose and justify a counting rule; include water routing, access, and overflow or equipment-failure fallback. Engineering performance checks are an extension after the core estimate.

PLANNER: _____ **TEAM RATE:** _____

CLASSROOM EXERCISE — NOT A REAL DEPLOYMENT ORDER

REQUEST 38 / KEEP THE DIRTY WATER OUT

ZOMBIE INVASION PREP / EMERGENCY RESPONSE EXERCISE

CLASSROOM SIMULATION • OPERATIONS REQUEST

SITUATION

Cleanup crews have isolated a contaminated loading pad. A new storm is about to send clean hillside runoff into it, multiplying the volume responders must collect.

REQUEST

Plan a clean-water diversion outside the exclusion zone and request support for a lined washwater collection pad inside the responder work area.

SITE INFORMATION

Clean diversion route is 36 ft with two 8 ft returns. Separate washwater pad inside dimensions: 10 × 8 ft at 0.25 ft operating depth. Washwater production is 2 gal/min for 60 minutes. Fill staging is 150 ft away on an approved clean route.

REQUESTED TIME WINDOW

Complete construction within 3 hours of assignment. Delivery, setup, inspection and finishing count. State weather/daylight assumptions and choose your planning factor.

OPERATIONAL CONSTRAINT

Students estimate resources; responders deploy inside the exclusion zone and choose compatible liner and recovery method. Clean runoff must reach approved drainage. Keep contaminated collection and drinking-water supplies separate.

ESTIMATE TO SUBMIT

Use the existing worksheet and your measured rate: bags and fill; time with 1 team and 4 teams; teams, people, equipment and worker support to meet the deadline. Choose and justify a counting rule; include water routing, access, and overflow or equipment-failure fallback. Engineering performance checks are an extension after the core estimate.

PLANNER: _____ **TEAM RATE:** _____

CLASSROOM EXERCISE — NOT A REAL DEPLOYMENT ORDER

REQUEST 39 / THE REPAIR WINDOW IS CLOSING

ZOMBIE INVASION PREP / EMERGENCY RESPONSE EXERCISE

CLASSROOM SIMULATION • OPERATIONS REQUEST

SITUATION

A shutoff valve in a shallow service yard is leaking. The technician can fix it if the immediate work area stays dry, but runoff will return when the next shower arrives.

REQUEST

Plan a small surface repair pocket, pumping and discharge protection; give the technician a usable work window.

SITE INFORMATION

Proposed U pocket: 12 ft front and two 8 ft returns. Separately specified discharge apron has 24 ft of ground-cover perimeter. Fill staging is 60 ft away. Scenario pump capacity: 6 gal/min; seepage: 2 gal/min. Initial removable water is 40 gallons.

REQUESTED TIME WINDOW

Complete construction within 2 hours of assignment. Delivery, setup, inspection and finishing count. State weather/daylight assumptions and choose your planning factor.

OPERATIONAL CONSTRAINT

Request liner, pump, screened intake and erosion-control apron. No excavation, trenches or underground entry. Route pumped water to approved drainage. Pumping time begins after installation; maintain access and electrical controls.

ESTIMATE TO SUBMIT

Use the existing worksheet and your measured rate: bags and fill; time with 1 team and 4 teams; teams, people, equipment and worker support to meet the deadline. Choose and justify a counting rule; include water routing, access, and overflow or equipment-failure fallback. Engineering performance checks are an extension after the core estimate.

PLANNER: _____ **TEAM RATE:** _____

CLASSROOM EXERCISE — NOT A REAL DEPLOYMENT ORDER

REQUEST 40 / THE DARK-SKY AID STATION

ZOMBIE INVASION PREP / EMERGENCY RESPONSE EXERCISE

CLASSROOM SIMULATION • OPERATIONS REQUEST

SITUATION

A mobile aid station must stay open through sunset. Its radios, charging tent and ground-mounted solar unit all need setup, while runoff is heading for the equipment area.

REQUEST

Plan bags and staffing for approved equipment ballast, ground-cover anchors and shallow runoff diversion before patient arrivals.

SITE INFORMATION

Approved tent plan: 8 stations × 4 bags. Approved low solar-frame plan: 6 stations × 3 bags. Ground supply cover: 12 × 8 ft, one bag per 4 ft of perimeter. Runoff line: 30 ft plus two 5 ft returns. Fill staging is 280 ft away. Tasks can use separately staffed work zones.

REQUESTED TIME WINDOW

Complete construction within 2 hours of assignment. Delivery, setup, inspection and finishing count. State weather/daylight assumptions and choose your planning factor.

OPERATIONAL CONSTRAINT

Request manufacturer-specified frames, anchors and attachments, lighting and safe power connections. Bag counts do not establish a wind rating. Keep routes accessible and protect supplies on approved platforms.

ESTIMATE TO SUBMIT

Use the existing worksheet and your measured rate: bags and fill; time with 1 team and 4 teams; teams, people, equipment and worker support to meet the deadline. Choose and justify a counting rule; include water routing, access, and overflow or equipment-failure fallback. Engineering performance checks are an extension after the core estimate.

PLANNER: _____ **TEAM RATE:** _____

CLASSROOM EXERCISE — NOT A REAL DEPLOYMENT ORDER

REQUEST 41 / THE LAST DRY SUPPLY LANE

ZOMBIE INVASION PREP / EMERGENCY RESPONSE EXERCISE

CLASSROOM SIMULATION • OPERATIONS REQUEST

SITUATION

Relief supplies arrive in two hours. The receiving area is wet, and a muddy path separates the delivery point from the shelter. Food boxes cannot sit in runoff while workers argue about where to put them.

REQUEST

Plan shallow diversion, anchored ground fabric and a lined boot-cleaning tray to keep a dry supply route operating.

SITE INFORMATION

Diversion route: 28 ft plus two 6 ft returns. Ground-fabric path: 24 × 4 ft; use one bag per 4 ft of perimeter. Boot tray inside dimensions: 6 × 4 ft at 0.25 ft operating depth. Fill staging is 160 ft away; a 6 ft accessible entrance must stay clear.

REQUESTED TIME WINDOW

Complete construction within 2 hours of assignment. Delivery, setup, inspection and finishing count. State weather/daylight assumptions and choose your planning factor.

OPERATIONAL CONSTRAINT

Request traction surface, approved decking/pallets, ground fabric, tray liner, brushes and washwater containers. Bags anchor edges outside walking paths; bags themselves are not stepping stones or deck supports.

ESTIMATE TO SUBMIT

Use the existing worksheet and your measured rate: bags and fill; time with 1 team and 4 teams; teams, people, equipment and worker support to meet the deadline. Choose and justify a counting rule; include water routing, access, and overflow or equipment-failure fallback. Engineering performance checks are an extension after the core estimate.

PLANNER: _____ **TEAM RATE:** _____

CLASSROOM EXERCISE — NOT A REAL DEPLOYMENT ORDER

REQUEST 42 / FOUR CALLS, ONE CREW

ZOMBIE INVASION PREP / EMERGENCY RESPONSE EXERCISE

CLASSROOM SIMULATION • OPERATIONS REQUEST

SITUATION

Dispatch receives four urgent calls at once: the radio needs power, toilets need flushing water, muddy runoff threatens the drain, and the aid tent cannot open. Every caller says their job comes first.

REQUEST

Submit one coordinated plan for four work zones, with priorities, dependencies and a defensible reduced-service fallback.

SITE INFORMATION

Radio U intake: 8 ft front + two 4 ft returns. Flushing basin: 10 × 6 ft inside at 0.25 ft depth. Mud diversion: 24 ft + two 6 ft returns. Approved aid-tent ballast: 6 stations × 4 bags. Fill staging is central; routes to the four zones are 60, 180, 300 and 420 ft respectively.

REQUESTED TIME WINDOW

Complete construction within 3 hours of assignment. Delivery, setup, inspection and finishing count. State weather/daylight assumptions and choose your planning factor.

OPERATIONAL CONSTRAINT

Initial card gives no supplied crew or stock: calculate and request them. A technician supplies the power-system design; responders approve water routes and tent plan. Preserve shelter access. Explain which jobs can proceed in parallel.

ESTIMATE TO SUBMIT

Use the existing worksheet and your measured rate: bags and fill; time with 1 team and 4 teams; teams, people, equipment and worker support to meet the deadline. Choose and justify a counting rule; include water routing, access, and overflow or equipment-failure fallback. Engineering performance checks are an extension after the core estimate.

PLANNER: _____ **TEAM RATE:** _____

CLASSROOM EXERCISE — NOT A REAL DEPLOYMENT ORDER